WeResence.

OFFICIAL PRD • V1.0

WeResence — Product Requirements Document (PRD)

Sovereign Dual-Transport Hybrid Messenger & Zero-Knowledge AI Ecosystem

DOCUMENT ID	**RES-WERESENCE-OFFICIAL PRD**	TARGET ARCHITECTURE	Android 8.0+ (API 26–34) / Progressive Web
LEAD ARCHITECT	**Piyush Singh** (Founder, Resence Ecosystem)	SECURITY CLASSIFICATION	**CONFIDENTIAL / PROPRIETARY SPECIFICATION**
RELEASE VERSION	v1.0.0 Production Blueprint	ECOSYSTEM DOMAIN	`resence.in` / `weresence.resence.in`

Product Requirements Document (PRD v1.0)

Project: WeResence (Sovereign Dual-Transport Hybrid Messenger & Private AI)

Organization: Resence Ecosystem (`resence.in`)

Author: Piyush Singh (Founder & Chief Architect)

Classification: Confidential / Proprietary Engineering Specification

Target Platforms: Android 8.0+ (API Level 26+) & Responsive Web Portal

Document Status: Approved / Production Blueprint

Executive Summary

WeResence is a next-generation sovereign communication platform engineered to solve the single greatest vulnerability of modern mobile messaging: **total dependency on active IP/cellular data connectivity and centralized surveillance backdoors.**

While industry leaders like Signal, Telegram, and WhatsApp offer varying degrees of security, they suffer catastrophic failures in adversarial environments:
1. When mobile data is turned off, throttled, or disconnected during network outages, transit dead zones, or

state-mandated internet shutdowns, all communication instantly halts.
2. Voice calling leaves clear digital footprints in the native Android `CallLog.Calls` provider, exposing social graphs to forensic physical inspection.
3. Screen recording and screenshot captures allow recipient-side data exfiltration without sender consent.
4. Shared galleries allow either party to delete evidence or tamper with mutual archives arbitrarily.

WeResence introduces the world's first **Dual-Engine Hybrid Transport Messenger**:
- **Primary Tier (High-Speed Online)**: Real-time WebSockets over TLS 1.3 with rich media streaming, group communities, and integrated Resence AI companion.
- **Failover Tier (Stealth Cellular SMS)**: Automatic, sub-second fallback to binary point-to-point SMS over dedicated **Port 8901**. Messages travel as encrypted binary byte arrays that bypass the OS Default SMS inbox, completely invisible to casual snoops and telecom content inspectors.
- **Sovereign Privacy Shield**: Hardware-enforced `FLAG_SECURE` window shielding, automated zero-trace native call log scrubbing, mutual private vault galleries with independent dual-passwords, decoy ghost PIN workspaces, and a zero-key client proxy gateway to Resence AI.

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1. Problem Statement & Market Opportunity

1.1 The Fragility of Modern IP Messengers

Every conventional messaging platform (WhatsApp, Telegram, Signal, iMessage) operates exclusively over the TCP/IP stack. If an LTE/5G or Wi-Fi data packet cannot be routed:
- Messages queue indefinitely in an "unsent" local state.
- Users are forced to drop back to unencrypted, plain-text carrier SMS, exposing personal conversations, phone numbers, and metadata directly to telecom operators, ISP surveillance, and third-party SMS reader apps.

1.2 The Forensic Leak Surface on Android

Even if a chat app encrypts data in transit:
- **Screenshot & Recording Vulnerability**: Default Android activities allow any recipient or background spyware to screenshot or record conversations without notifying the sender.
- **System Call Log Footprint**: VoIP calls routed through Android's standard telecom framework frequently leak caller identity, timestamps, and call durations into the stock dialer history.
- **Physical Device Coercion**: If an adversary compels a user to unlock their phone, all private chats and media are immediately exposed unless a mathematically isolated decoy environment exists.

1.3 The WeResence Value Proposition

WeResence bridges high-speed digital communications with ultra-resilient cellular radio signaling, wrapping the entire experience in a zero-knowledge hardware-backed security envelope that looks, feels, and performs like a tier-one consumer application.

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2. User Personas & Target Demographics

Persona 1: The Privacy-Conscious Campus Student

- **Profile**: 18–24 years old, college student (e.g. GLBIM Greater Noida, Delhi NCR).
- **Environment**: High campus density, spotty basement Wi-Fi, shared mobile hotspots, strict parental or peer curiosity.
- **Core Pain Point**: Wants to chat freely with trusted friends and share notes/media without worrying about someone taking screenshots, peeking over shoulders, or seeing chat records in dialer histories.
- **WeResence Utility**: Uses Mutual Vault with independent passwords, Ghost PIN decoy workspace, and automatic offline SMS switching during college connectivity blackouts.

Persona 2: The Field Professional & Rural Commuter

- **Profile**: 25–45 years old, field engineer, researcher, or frequent traveler across rural rail and highway corridors.
- **Environment**: Severe 4G/5G edge drops, high packet loss, battery conservation necessity.
- **Core Pain Point**: Important 1-on-1 operational updates get delayed because WhatsApp/Telegram requires high-bandwidth data handshake.
- **WeResence Utility**: Zero-drop auto-shift keeps critical 1-on-1 coordination moving smoothly over raw cellular SMS without requiring user intervention.

Persona 3: The Investigative Journalist / High-Security Operator

- **Profile**: 30–60 years old, handles sensitive investigations, high risk of state or corporate surveillance.
- **Environment**: Hostile networks, active IMSI-catchers, potential device seizure.
- **Core Pain Point**: Centralized servers store metadata logs; plain SMS is logged by telecom operators in cleartext.
- **WeResence Utility**: Port 8901 binary encryption makes SMS payloads look like random binary noise. Android Keystore StrongBox enclave protects local SQLCipher databases.

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3. Competitive Analysis & Feature Matrix

Capability	WeResence v1.0	Signal	Telegram	WhatsApp	Session
Offline Transport (No Internet)	**Yes (Port 8901 SMS)**	No	No	No	No
Automatic Online ↔ Offline Shift	**Yes (< 500ms)**	No	No	No	No
Carrier Inbox Bypass (Stealth)	**Yes (0% SMS Leak)**	No	No	No	No
End-to-End Encryption	**Double Ratchet (Curve25519)**	Signal Protocol	Secret Chats Only	Signal Protocol	Onion Routing
Anti-Screenshot Hardware Shield	**Yes (`FLAG_SECURE`)**	Opt-in	Secret Chats Only	View Once Only	Opt-in
Automated Native Call Log Scrub	**Yes (Zero Trace)**	No	No	No	No
Mutual Private Vault Gallery	**Yes (Dual Password)**	No	No	No	No
Decoy Ghost PIN Workspace	**Yes (`0000` vs `1234`)**	No	No	No	No
Built-in AI Companion	**Yes (@ResenceAI)**	No	No (Third-party bots)	Meta AI (Data Logged)	No
Offline AI Freeze Protection	**Yes (Quota Guard)**	N/A	N/A	No	N/A
Phonebook Matching with Viral Loop	**Yes (Hashed Sync)**	Yes	Yes	Yes	No

4. System Architecture & Core Engine Modules

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ARCHITECTURE FLOW DIAGRAM
```

```
flowchart TD
    subgraph UI ["Presentation Layer (Android & Web)"]
        A1[1-on-1 Chat Screen]
        A2[Communities / Groups]
        A3[Resence AI Tab]
        A4[Mutual Vault Gallery]
        A5[Decoy PIN Gatekeeper]
    end

    subgraph ENGINE ["Core Connection Engine"]
        B1[Live Network Observer]
        B2{Data Connection Active?}
        B3[Primary Online Transport: WebSockets / TLS 1.3]
        B4[Failover Stealth Transport: Port 8901 Binary SMS]
    end

    subgraph SECURITY ["Zero-Knowledge Hardware Shield"]
        C1[Android Keystore StrongBox]
        C2[SQLCipher 256-bit DB]
        C3[GPU SurfaceFlinger FLAG_SECURE]
        C4[CallLog.Calls Scrubbing Daemon]
    end

    subgraph CLOUD ["Resence Cloud & AI Infrastructure"]
        D1[Online WebSocket Relays]
        D2[Resence AI Zero-Key Gateway]
        D3[Pooled LLM Engine: Gemini 2.5 + Groq]
    end

    UI ⟶ B1
    B1 ⟶ B2
    B2 -- "Yes (Wi-Fi / 5G)" ⟶ B3
    B2 -- "No (Data Drop)" ⟶ B4
    B3 ⟷ D1
    B3 ⟷ D2
    D2 ⟷ D3
    B3 ⟶ SECURITY
    B4 ⟶ SECURITY
```

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5. Functional Requirements & Feature Specifications

5.1 Dual-Transport Engine (Online WebSockets 🔄 Offline Port 8901 SMS)

- **FR-1.1 Automatic Network Sensing**: The application must monitor active network capabilities via Android's `ConnectivityManager.NetworkCallback`. When `NET_CAPABILITY_INTERNET` is validated, the app operates in **Online Mode**.
- **FR-1.2 Zero-Drop Failover**: When internet connectivity is lost for greater than 1,500 milliseconds during an active 1-on-1 conversation, the transport pipeline must automatically shift to **Stealth Cellular SMS Mode**.

- **FR-1.3 Sub-Second Recovery**: Once internet connectivity is re-established and verified via HTTP probe, queued outbox messages and incoming streams must smoothly transition back to WebSockets without UI interruption.
- **FR-1.4 Scope Boundary**:
- **1-on-1 Chats**: Support both Online WebSockets and Offline SMS.
- **Communities & Group Channels**: **Strictly 100% Online**. Communities must NEVER attempt to route over SMS to prevent exponential carrier SMS billing and broadcast flooding.
- **FR-1.5 Daily SMS Quota Counter**:
- The UI must maintain an on-device daily counter tracking cellular SMS consumption against the user's carrier limit (default: 100 SMS/day).
- When the quota reaches 100, the app must pause automatic cellular transmission and queue messages in an encrypted local outbox until data returns or the next calendar day begins.

5.2 Stealth Cellular Messaging (Port 8901 Binary SMS)

- **FR-2.1 Port-Directed Addressing**: Offline messages must be dispatched using Android's native `SmsManager.sendDataMessage()` targeting destination **Port 8901**.
- **FR-2.2 OS Inbox Isolation**: The app must register a high-priority `BroadcastReceiver` (`priority = 999`) listening specifically for `android.intent.action.DATA_SMS_RECEIVED` on port 8901.
- **FR-2.3 Zero Carrier Inbox Leak**: Incoming packets on Port 8901 must be consumed directly by WeResence and aborted from the standard broadcast chain (`abortBroadcast()`). The standard Android Google Messages or Samsung Messages application must receive zero notifications, zero unread alerts, and zero thread entries.
- **FR-2.4 Binary Serialization**: Payloads must be compacted into binary Protocol Buffers (Protobuf) or compact binary byte arrays before encryption, maximizing the 133-byte single-segment SMS payload limit.
- **FR-2.5 Multi-Segment Reassembly**: Messages exceeding a single SMS frame must be split with a 4-byte header (`[MsgID:2][Seq:1][Total:1]`) and reassembled on the recipient device before decryption.

5.3 Sovereign Privacy & Anti-Surveillance Suite

- **FR-3.1 Hardware Anti-Screenshot (`FLAG_SECURE`)**:
- All Android Activity windows must declare `WindowManager.LayoutParams.FLAG_SECURE` .
- Android system screenshots (`Power + Vol Down`), Google Assistant screen captures, third-party screen recorders, and Android task-switcher app snapshots must be blocked at the OS surface level, rendering completely black frames.
- **FR-3.2 Native Call Log Scrubbing Engine**:
- When a WeResence voice call terminates, a background daemon must monitor Android's `CallLog.Calls` provider via `ContentObserver` .
- Any call entry matching the session peer's phone number or WeResence VoIP component must be surgically deleted within 300 milliseconds.
- The stock Android phone dialer must exhibit a 0% forensic record of the call having occurred.

- **FR-3.3 Decoy Ghost PIN Workspace**:
- The application must support two distinct PINs:
  - **Master PIN (`1234`)**: Unlocks the user's authentic sovereign workspace, real chat history, and true mutual vaults.
  - **Decoy Ghost PIN (`0000`)**: Unlocks an entirely benign, fabricated workspace with simulated contacts, generic chats, and empty vaults.
- Under physical coercion, entering the Decoy PIN provides total plausible deniability without wiping or damaging the authentic database.

5.4 Mutual Private Vault Gallery & Lossless Media Pipeline

- **FR-4.1 Dual-Password Architecture**:
- Each 1-on-1 relationship features a dedicated, shared encrypted media gallery.
- Access to the gallery requires independent PIN/passwords set by each user. User A's password does not grant access on User B's device; each derives independent decryption keys.
- **FR-4.2 Uploader-Only Deletion Governance**:
- Photos, videos, and documents stored inside the Mutual Vault are tagged with an immutable cryptographic signature of the uploader.
- The non-uploading peer has read, view, and zoom permissions inside the secure viewport.
- **The delete button is cryptographically restricted to the original uploader.** The non-uploading recipient CANNOT delete or tamper with files uploaded by their partner.
- **FR-4.3 Anti-Exfiltration Viewport**:
- Media inside the Mutual Vault cannot be exported to the public Android gallery, cannot be forwarded, and cannot be captured via screenshot (`FLAG_SECURE`).
- **FR-4.4 Zero-Compression Lossless Media Transmission (Anti-WhatsApp Degradation)**:
- **The WhatsApp Compression Flaw**: WhatsApp aggressively recompresses photos with lossy JPEG at ~70% quality, downscaling high-res images to 1080p, and transcodes videos down to 480p/720p at low bitrates, causing blurriness, banding, and artifacting.
- **WeResence Lossless Bitstream Transmission**: All photos (RAW, PNG, HEIC, 48MP/108MP) and 4K 60FPS videos are transmitted in their **100% UNCOMPRESSED ORIGINAL BITSTREAM**.
- **Cryptographic Checksum Verification**: Every file generates a pre-transmission SHA-256 hash. The recipient verifies `Hash(Sent) == Hash(Received)`, guaranteeing zero loss of fidelity.
- **Lag-Free 120 FPS Fluidity**: Direct GPU `HardwareBitmap` memory pipeline bypassing CPU RAM, preventing garbage-collection micro-stutters during high-speed feed scrolling.

5.5 Phone-Number Anchored Identity & Organic Viral Loops

- **FR-5.1 Dual-Identity Model**:
- **Private Anchor**: E.164 verified phone number (e.g. `+91 98765 43210`) used for carrier SMS routing and initial cryptographic handshake.

- **Public Display**: Unique sovereign username handle (e.g. `@piyush`) used for campus communities, group chats, and public discovery.
- **FR-5.2 Zero-Knowledge Phonebook Sync**:
- Device contacts must be hashed using salted SHA-256 before comparing against the directory server.
- Raw phonebooks are NEVER uploaded or stored on any server.
- Contacts already registered on WeResence are automatically tagged with a green "Available on WeResence" badge.
- **FR-5.3 One-Click Viral Invite Loop**:
- Unregistered contacts feature an immediate "Invite" action.
- Tapping invite generates a pre-formatted invitation SMS containing a cryptographic referral link: `"Hey Rahul, join me on WeResence! Chat privately with anti-screenshot protection, even without internet via offline SMS: https://weresence.resence.in/invite/piyush"`

5.6 Built-in Resence AI Companion (`@ResenceAI`)

- **FR-6.1 Default Pinned Assistant**:
- `@ResenceAI` is pinned at the top of the chat list for every new install, alongside a dedicated bottom navigation tab.
- **FR-6.2 Zero-Key Client Architecture**:
- The APK client binary must NEVER contain raw Gemini or Groq developer API keys.
- All AI queries route through the secure proxy gateway `https://api.resence.in/v1/ai/chat` using short-lived device session tokens.
- **FR-6.3 Intelligent Multi-Model Routing**:
- Standard text queries and voice prompts route to **Groq Llama 3.3 70B** (< 300ms latency, 500 tokens/sec).
- Images, photos, and document attachments route automatically to **Google Gemini 2.5 Flash Multimodal** for visual OCR, diagram analysis, and deep reasoning.
- **FR-6.4 Strict Offline Quota Guard**:
- When the device is offline (cellular SMS mode), `@ResenceAI` companion **automatically freezes**:
  > *"Resence AI requires active data to think. Offline SMS is reserved exclusively for sovereign 1-on-1 human communication."*
- This prevents AI replies from consuming the user's daily carrier SMS quota.

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6. Information Architecture & Screen-by-Screen UX

```
ARCHITECTURE FLOW DIAGRAM

graph TD
    Splash[Splash & Security Initializer] ⟶ Onboarding[Phone Verification & Keystore Gen]
    Onboarding ⟶ PinSetup[Set Master PIN & Optional Ghost PIN]
    PinSetup ⟶ MainApp[Main Application Hub]

    subgraph "Main Application (4-Tab Interface)"
        MainApp ⟶ Tab1[Tab 1: 1-on-1 Chats Feed]
        MainApp ⟶ Tab2[Tab 2: Communities & Groups]
        MainApp ⟶ Tab3[Tab 3: Resence AI Companion]
        MainApp ⟶ Tab4[Tab 4: Settings & Security Vault]
    end

    Tab1 ⟶ ChatRoom[1-on-1 Secure Chat View]
    ChatRoom ⟶ NetIndicator[Live Network Status: Online / Offline SMS]
    ChatRoom ⟶ ProfileModal[Peer Contact Profile]
    ProfileModal ⟶ MutualVault[Mutual Private Vault Gallery]

    Tab2 ⟶ CommunityHub[Campus Community: e.g. GLBIM Greater Noida]
    CommunityHub ⟶ Channels[Announcements, Sub-Groups, Notes]

    Tab3 ⟶ AiChat[Real-Time Streaming AI Assistant]
    AiChat ⟶ MultimodalUpload[Photo / Document Breakdown]

    Tab4 ⟶ VaultManager[Vault PIN & Biometrics Config]
    Tab4 ⟶ SimManager[Offline SMS SIM Preference & Counter]
    Tab4 ⟶ GoogleLink[Linked Google Profile: Optional Backup]
```

6.1 Screen 1: Splash & Hardware Attestation

- Initializer verifies Android Keystore availability, initializes SQLCipher native binaries, sets `FLAG_SECURE` on the window, and checks Root/Jailbreak status.

6.2 Screen 2: Phone Verification & Identity Setup

- Clean, minimal OLED dark screen.
- User enters country code and phone number.
- Automatic SMS OTP detection (or silent cellular token verification).
- User picks their unique public handle (`@username`) and sets their profile avatar.

6.3 Screen 3: The 1-on-1 Chat Interface

- **Dynamic Header**: Shows peer avatar, display name, public handle, and real-time transport pill:
- 🟢 `ONLINE` : Connected via WebSockets. Full media attachment available.
- 🟡 `OFFLINE (PORT 8901)` : Switched to binary SMS. Shows daily SMS counter (`14/100`). Media attachment icon disabled with tooltip.
- **Message Bubbles**:
- Sent Online: Dark emerald bubble with double cyan checkmarks (`✓✓ Delivered`).

- Sent Offline SMS: Subtle dark teal bubble with radio wave icon and single checkmark (`✓ Sent via Port 8901 SMS`).
- **Encryption Pill**: Subtly displays `🔒 Zero-Knowledge E2EE • Anti-Screenshot Active` at the top of the feed.

6.4 Screen 4: The Mutual Private Vault

- Accessed by tapping peer profile → **Mutual Private Gallery**.
- Prompts for the user's custom vault password.
- Displays encrypted grid of photos and documents:
- Items uploaded by user feature an active **Delete** button.
- Items uploaded by peer display **View Only** badge (no delete option).
- Floating Action Button: `➕ Add Media to Mutual Vault` .

6.5 Screen 5: Campus Communities & Sub-Groups

- List of university/college communities (e.g. `GLBIM Greater Noida`).
- Sub-divided into structured channels:
- `#Announcements` (Admin-only broadcast)
- `#BCA AI/ML Sem 3` (Collaborative peer chat)
- `#Practical Code & Notes` (Document and code snippets)
- Always operates 100% online; clearly notifies user if disconnected.

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7. Non-Functional Requirements (NFRs)

7.1 Performance & Latency SLAs

- **App Cold Start**: < 800 milliseconds on mid-tier Android devices (Snapdragon 680 / Helio G99).
- **Online Message Delivery**: < 200 milliseconds (p95) under 4G/5G/Wi-Fi.
- **Offline SMS Failover Time**: < 1,500 milliseconds from loss of IP carrier packet to ready SMS transmission state.
- **UI Render Framerate**: Rock-solid 60 FPS / 120 FPS on high-refresh OLED displays; zero jank or frame dropping during list scrolls.

7.2 Storage & Memory Footprint

- **APK Package Size**: ≤ 25 MB compressed download size.
- **Runtime RAM Utilization**: ≤ 75 MB idle, ≤ 140 MB peak during multimodal AI streaming.
- **Local Database Growth**: Encrypted SQLite tables pruned of ephemeral handshake cache after 30 days.

7.3 Battery & Radio Efficiency

- **Foreground Drain**: < 2.5% battery consumption per 1 hour of active messaging.
- **Background Standby**: < 0.8% battery drain per 24 hours. Zero aggressive background wake locks; network polling handled strictly via OS-native `ConnectivityManager` callbacks.

7.4 Security & Compliance Standards

- **Cryptographic Primitives**: Strictly NIST / FIPS approved algorithms (AES-256-GCM, HMAC-SHA256, Curve25519 / Ed25519).
- **Zero Local Plaintext**: No unencrypted chats, session keys, or media files stored on flash storage.
- **ProGuard / R8 Obfuscation**: 100% of internal classes, method signatures, and cryptography wrappers obfuscated with aggressive name mangling.

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8. Release Roadmap & Milestones

Phase 1: MVP Core (v1.0 – Current Production Release)

- [x] Web landing page and brand assets finalized in 100% OLED Dark Mode.
- [x] Dual-engine network switcher (`Online WebSockets` 🔄 `Port 8901 SMS`).
- [x] Hardware `FLAG_SECURE` anti-screenshot and recording block.
- [x] Native `CallLog.Calls` automated scrubbing engine.
- [x] Decoy Ghost PIN (`0000`) and Master PIN (`1234`) architecture.
- [x] Mutual Private Gallery with uploader-only deletion control.
- [x] Zero-key client proxy for Resence AI (Gemini 2.5 + Groq).
- [x] Formal PRD and TRD documentation compiled into vector PDFs.

Phase 2: On-Device Intelligence & WebRTC (v1.5 – Q3 2026)

- [ ] On-device **Gemma 2B Nano (GGUF)** integration via Google MediaPipe for true offline, zero-data AI reasoning.
- [ ] Direct P2P WebRTC end-to-end encrypted voice & video calling.
- [ ] Automated self-destruct timers (ephemeral disappearing messages from 5 seconds to 7 days).

Phase 3: Mesh Networking & Sovereign Identity (v2.0 – Q4 2026)

- [ ] Bluetooth Low Energy (BLE) & Wi-Fi Direct multi-hop mesh networking for zero-carrier, zero-internet local campus communications.
- [ ] Decentralized DID (Decentralized Identity) support for SIM-less anonymous account creation.

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9. Success Metrics & Telemetry (Privacy-Preserving)

To preserve sovereign user privacy, WeResence operates **Zero User Content Telemetry**. Metrics are strictly aggregated on-device and reported as anonymized differential counts:

1. **Failover Efficiency Rate**: Percentage of network drop events that successfully deliver the queued message over Port 8901 SMS without manual retry (Target: > 98.5%).
2. **Crash-Free Sessions**: Stability index across diverse Android OEM skins (MIUI, OneUI, ColorOS, Stock) (Target: > 99.8%).
3. **Viral Invite Conversion (K-Factor)**: Average number of successful app installs generated per SMS invite link shared (Target: $K \geq 1.4$).
4. **Offline Quota Adherence**: 0% carrier bill shock reports; zero cases of messages exceeding the daily carrier limit.